6 · Investigate a paving trend

SKILLEDTRADESBC · LINE A

A6 · Describe future paving/construction trends

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A6.1 · Describe green roads	LA6-01, LA6-04
A6.2 · Describe new paving materials	LA6-03–04
A6.3 · Describe new sub-bases	LA6-02
A6.4 · Describe floating roads	LA6-02, LA6-04
A6.5 · Describe added value enhancements	LA6-03, LA6-06

Turn a promising idea into a testable question

A trend is useful to investigate when it responds to a defined need. A product described as *green*, *new* or *innovative* still needs evidence. Ask what problem it addresses, how it works, where it has been studied, what limits apply and who would approve its use. Something new to a local project may have been used elsewhere for decades. Publication date and location matter.

This unit compares material and design ideas without selecting a live road design. You will prepare a recommendation for further investigation using complete evidence cards and a fictional project brief. A careful recommendation can say that one option deserves a closer technical study while leaving design approval to the appropriate professional.

Keep the outline term and explain the mechanism

Outline wording	Meaning to investigate in this unit	First useful question
Green roads	A claim about an environmental purpose or outcome	Which outcome, comparison and time period?
Pave stones / paving stones	Individual paving units in a designed system	Does this particular system manage water through its joints?
Porous asphalt	A surface with interconnected voids, used within an appropriate pavement system	Where will water go, and how will the system be maintained?

Outline wording	Meaning to investigate in this unit	First useful question
Modified asphalt	Asphalt whose binder or mixture has been changed for a defined purpose	Which property changes, and what evidence supports the use?
New sub-bases / pumice	Foundation or fill ideas whose actual design role must be established	Is this material proposed as sub-base, embankment fill or something else?
Floating roads / Styrofoam	Historical outline wording discussed here through engineered lightweight fill and EPS geofoam	What engineered material and geotechnical design are intended?
Street print	A patterned surface treatment or visual enhancement	What are its purpose, performance and maintenance requirements?

A trade term helps you find the right subject. The project documents establish the actual material and use. Keep *sub-base* and *embankment fill* distinct when interpreting the historical cost example below.

Compare surface and reuse approaches

Trend card N1 · Paving stones. Modular paving uses individual units rather than one continuous asphalt layer. Some systems are designed to manage water through joints and underlying layers; the word *paver* alone does not establish permeability. Ask about the whole pavement system, loading, drainage, accessibility and maintenance. A patterned appearance is not environmental evidence.

Trend card N2 · Porous asphalt. FHWA's April 2015 technical brief describes porous asphalt pavements with stone reservoirs as systems intended to provide a surface and manage stormwater. Water moves through interconnected surface voids into the underlying reservoir. This differs from an open-graded friction course on a dense pavement, where water moves through the surface course and out toward the side. The brief identifies potential clogging and the need for specialized maintenance, as well as site/design limitations. A material name does not guarantee infiltration at an unsuitable site.

Trend card N3 · Recycling. Recovered asphalt may enter an approved plant or specialist in-place process. Recovered concrete may be processed as aggregate. Potential reductions in new-material demand need to be examined alongside processing, transport, suitability, maintenance and end-of-life. “Recycled” is a property of the material pathway, not a complete environmental score.

LA6-01 · Improve three claims

Rewrite these statements so they are bounded and useful: “All paving stones absorb rain”; “Porous asphalt never needs drainage planning”; “Any recycled material is automatically the lowest-impact choice.” For each, identify a mechanism, a condition and one question for a qualified reviewer. **Skills:** Reading, Problem Solving, Writing.

Ask what a material changes

Trend card N4 · Epoxy, colour and modified asphalt. Epoxy systems, coloured surfaces and modified binders are families of products and applications. Colour can serve appearance or wayfinding, but it does not by itself establish structural performance, skid properties, durability or accessibility. A modified binder

may be selected for a particular performance requirement. The full product description, intended use, test evidence, compatible construction system and approved specification are needed before adoption. Do not turn a product family into one universal recipe.

Trend card N5 · Lightweight fill and "floating roads." Lightweight fill reduces the load imposed by an embankment compared with a heavier fill arrangement. Expanded polystyrene, or EPS, geofoam is one engineered lightweight-fill material. Its use beneath a road does not mean the road floats on open water. The soil, loading, stability, protection, groundwater and installation conditions require geotechnical design. A brand name or generic word such as *foam* is not a material specification. Pumice is another material encountered in lightweight-fill discussions, with properties and availability that must be checked for the actual source and use.

FHWA's **April 2017 Ground Modification Methods, Volume I** describes lightweight fills and EPS geofoam. It explicitly states that the publication is not a standard, specification or regulation. Its chapter explains that material properties vary and detailed design requires further testing and evaluation. The examples show that lightweight-fill ideas are not all recent inventions.

Evidence table N6 · One historical U.S. cost example. FHWA's 2017 manual reproduces a Milpitas, California project comparison credited to Cheng 2016. The total costs below include material, transport and longevity costs. They **exclude installation and contractor overhead and profit**. They are historical U.S. project figures, not current B.C. quotes. The table's dollar figures are retained in their source context; no currency conversion is supplied.

Material	Reported example total
Traditional soil fill	$563,000
Pumice rock	$633,000
EPS-block geofoam	$1,145,000
Expanded shale/clay	$490,000
Wood chips	$545,000
Tire-derived aggregate	$334,000

Worked example. In this table, pumice's reported total exceeds traditional soil fill by `$633,000 − $563,000 = $70,000`. That result answers a question about this historical comparison. It does not establish that pumice always costs $70,000 more, that the cheapest listed option meets a new site's design needs, or that the table includes every cost.

LA6-02 · Read the cost boundary

Calculate the difference between EPS-block geofoam and pumice in N6. Name three reasons the table cannot be used as a current B.C. quotation. Explain why a lightweight-fill cost table does not by itself establish an approved pavement sub-base material. Explain why price ranking alone cannot establish technical suitability. Keep units and the word *historical* in your conclusion. **Skills:** Numeracy, Reading, Problem Solving.

Separate an enhancement from a requirement

Trend card N7 · Patterned surfaces. Street-print or other patterned treatments can add a visual effect to a pavement. A visual enhancement may serve a project purpose, but performance claims still need evidence. Ask about the actual treatment, friction, wear, maintenance, access needs and compatibility with the underlying system. A decorative pattern is not evidence of structural strengthening or permission to use a cultural design. Any community-specific image or knowledge requires appropriate permission.

LA6-03 · Make a product enquiry

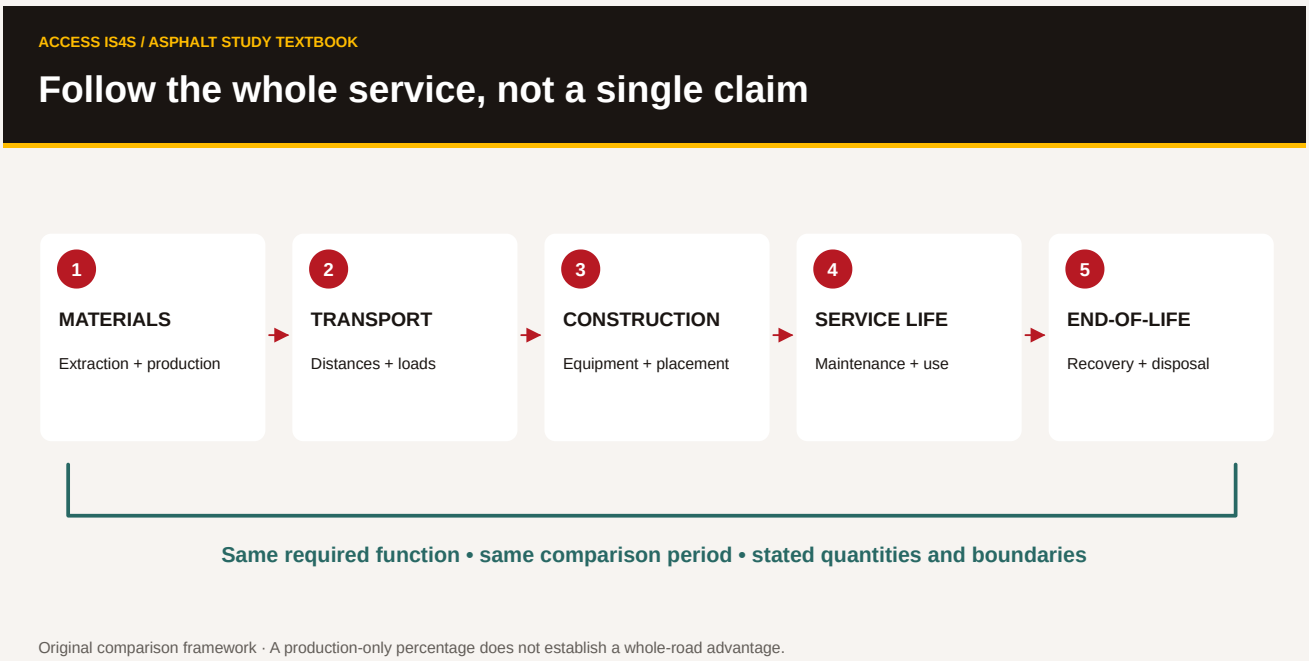
Choose N4 or N7 and draft five questions for a fictional supplier. Include intended use, product identity, evidence, maintenance and approval/permission. Make one question specific enough that a vague marketing answer would be recognizable. Keep the enquiry local; no message is sent. **Skills:** Writing, Communication, Creativity and Innovation.

Apply the evidence to a defined case

Fictional project N8 · A learning-centre courtyard. A facility team is considering a 600 m² courtyard used by pedestrians and occasional authorized service vehicles. The team wants to improve surface-water management and readable wayfinding. No geotechnical investigation, infiltration test, approved loading design, maintenance budget or cultural-design permission has been supplied. The project budget is not yet set. A promoter offers porous asphalt, coloured asphalt or an EPS-supported pavement as three “green” options. These options address different questions and are not interchangeable solutions.

FIGURE T19

Follow the whole service, not a single claim



Note. ACCESS IS4S. (2026). Follow the whole service, not a single claim [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

LA6-04 · Compare options against the actual need

Make a table for the three offers. State the problem each might address, evidence provided in N1–N7, missing project information and the reviewer needed. Select one or two ideas for further investigation, with reasons. Do not approve construction or claim a total environmental winner. Each learner contributes an evidence-based question before the group makes its shortlist. **Skills:** Problem Solving, Collaboration, Reading.

LA6-05 · Adapt to a new constraint

The fictional facility manager says there is no commitment to specialized pavement-cleaning equipment and asks for a maintenance plan before further consideration. Revise your shortlist or investigation plan. Explain why this new fact is relevant to porous asphalt and why it does not automatically approve another option. **Skills:** Adaptability, Communication, Problem Solving.

LA6-06 · Build an evidence brief and test its clarity

Create a one-page brief containing the project question, one proposed investigation, a supported benefit, a limitation, missing information and a next decision. Add source titles/dates from the research record and mark predictions as predictions. Save and reopen the file if using Digital. Ask a partner to identify what you are recommending and what you are leaving undecided; revise one unclear sentence. **Skills:** Writing, Digital, Communication, Creativity and Innovation.

Keep the claims you improved, historical comparison, supplier questions, option table and revised brief. A useful innovation record shows the question, evidence, trial or investigation needed and the limits of the conclusion.

7 · Recognize hazards and prepare for safe work

SKILLEDTRADESBC · LINE B

B1 · Safety orientation

Recognize the hazard, understand the control, confirm the responsibility and explain the next step.

B1 learning task	Classroom evidence
1 · Describe PPE: responsibilities, selection, use and care	TB7-01–02
2 · Describe workplace hazard assessment	TB7-03–04
3 · Describe company safety policies and orientation	TB7-05
4 · Review WHMIS and first-aid requirements	TB7-06–07
5 · Review WorkSafeBC rigging requirements	TB7-08–09
6 · Describe site-specific safety risks	TB7-10–11
7 · Describe public safety and equipment security	TB7-12–13

Begin with the work and the people affected

A **hazard** is a source or situation with the potential to cause harm. **Risk** considers the likelihood and consequences of that harm under the actual conditions. A **control** changes those conditions to prevent or reduce exposure. A control needs an owner, a way to check it and a response if it fails. “Be careful” does not tell a new worker which action is required.

The paving crew works around moving equipment, hot materials, changing surfaces and other people. A task that was controlled yesterday may change with wind, visibility, traffic, equipment condition or a different worker. Before work begins, the crew needs a shared understanding of the task and the current controls. During the work, a changed condition needs attention rather than silent improvisation.

The examples in this chapter teach recognition and decision making. The workplace must provide its own orientation, procedures, training, qualified supervision and emergency arrangements. Read the complete applicable WorkSafeBC requirements when preparing actual work. A classroom exercise does not qualify a person to rig a load, operate machinery, direct traffic or act as the designated first-aid attendant.

7.1 · PPE is selected for a hazard and a person

Personal protective equipment (PPE) is equipment worn to protect a person against specified hazards. Selection starts with an assessment of the exposure and the work. A product that protects against one hazard may not protect against another. A general-purpose glove is different from a glove selected for a particular chemical; eye protection and face protection have different roles; hearing protection must suit the assessed noise exposure and the required program.

Fit and compatibility matter. A respirator must be appropriate for the contaminant and conditions and used within the required respiratory-protection program. A generic “dust mask” description does not establish adequate protection against a gas, oxygen deficiency or a particular dust. A facepiece, eyewear, headgear and other items must work together without making one another ineffective. Training includes the limits of the equipment, not just how to put it on.

Know the provision responsibilities

WorkSafeBC OHS Regulation **8.2** makes the worker responsible for clothing needed for the natural elements, general-purpose work gloves, appropriate footwear including safety footwear, and safety headgear. The employer provides all other PPE required by the regulation at no cost to the worker. Agreements may provide for the employer to supply some or all of the worker-responsibility items. If employer-provided PPE causes allergenic or other adverse health effects, the employer must provide appropriate alternatives or safe measures. Read the actual agreement and the complete regulation before deciding how an item is supplied.

Responsibility to provide an item does not remove the employer's duties concerning suitable protection, instruction and the workplace. Under **8.3**, PPE must provide effective protection, be compatible, avoid creating a hazard and be maintained in good working order and sanitary condition. Where workplace evaluation is required, **8.4** addresses consultation. **8.5** specifies program elements for chemical exposure or oxygen-deficient atmospheres, including selection, use, inspection, cleaning, maintenance, storage and training.

Check	What to establish	Example of a useful record
Hazard and task	What exposure is being controlled?	“Splash task; actual product SDS and procedure identified.”
Selection	Does the equipment protect against this exposure?	Product/model, standard or rating where required, selection confirmation
Fit and compatibility	Can the person use it correctly with other PPE?	Fit/training record as applicable; unresolved interference reported
Inspection	Is it complete, serviceable and within its permitted use?	Defect or condition recorded; unsuitable equipment withdrawn
Use and limits	What can it protect against, and when is it insufficient?	Demonstration and explanation checked by the instructor/supervisor
Care and storage	How is contamination, damage or deterioration prevented?	Manufacturer procedure and assigned care arrangements

TB7-01 · Follow the responsibility without losing the hazard

Sort these fictional supply questions using 8.2: weather jacket, general-purpose work gloves, safety footwear, safety headgear, and chemical-specific protection required by the assessed task. Explain how an employer agreement could affect the first four. Then explain why “I bought it myself” does not establish that an item is suitable. Keep a written answer with the clause and the outstanding selection question. **Skills:** Reading, Writing, Problem Solving.

TB7-02 · Inspect the decision, not just the item

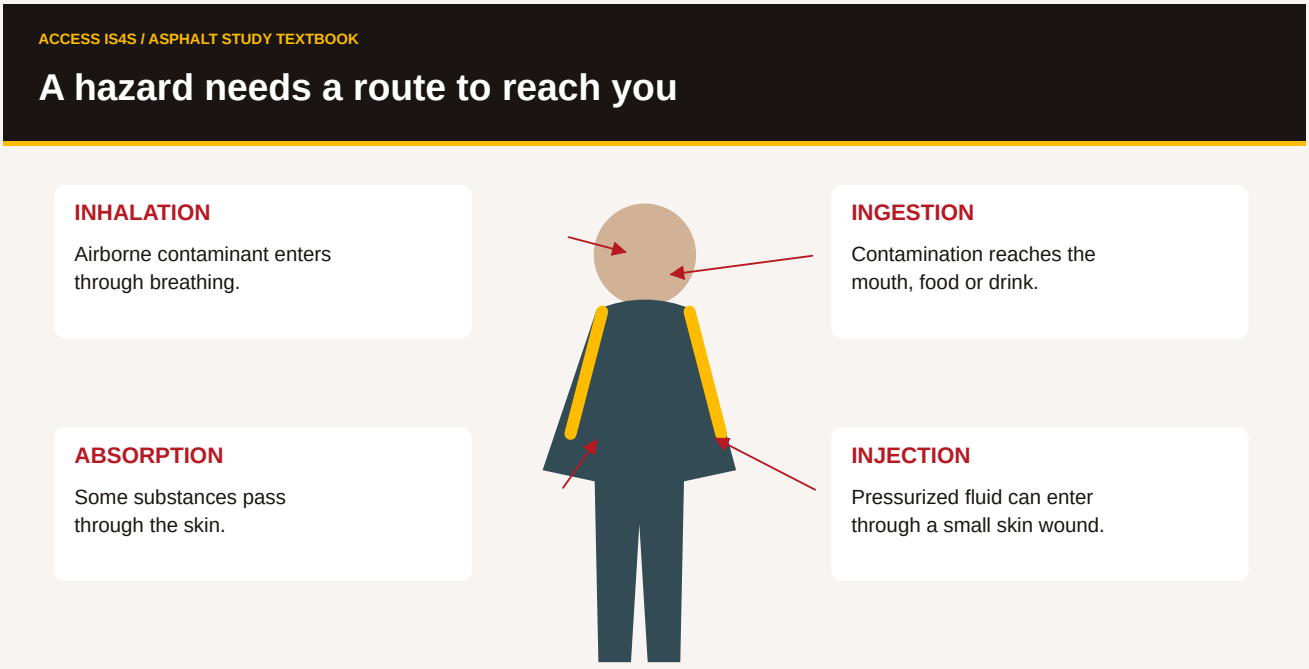
Fictional observations: a learner's eyewear lifts a respirator seal; a glove's chemical suitability is unknown; a headgear component is cracked; a reusable item is visibly contaminated. Make a four-row action record. Identify the unresolved protection issue, how exposure is avoided while it is resolved, who must assess it and what evidence is needed before use. Do not prescribe a substitute product. Explain one case aloud and invite a partner to check it. **Skills:** Communication, Collaboration, Problem Solving.

7.2 · Recognize how exposure happens

A hazard assessment identifies the job steps, people exposed, hazards, existing controls and further controls needed. It must consider routine work and foreseeable changes. Cleaning, clearing a blockage, maintenance, refuelling and a public-access change can introduce hazards that are absent from normal production.

FIGURE T20

A hazard needs a route to reach you



Note. ACCESS IS4S. (2026). A hazard needs a route to reach you [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Inhalation brings an airborne contaminant into the respiratory system. Carbon monoxide can occur with combustion equipment; hydrogen sulphide can be a concern in particular petroleum-related or confined-space conditions; welding produces fumes specific to the process and materials. Dust-generating work can create respirable crystalline silica exposure. Disturbing suspect asbestos-containing material requires identification and the applicable controls. These examples identify possible hazards to assess, not a claim that every paving site contains every substance. Smell, appearance and a photograph cannot establish a safe airborne concentration.

Ingestion can occur when contamination reaches the mouth through hands, food, drink or other objects. **Absorption** occurs when a substance enters through skin; its significance depends on the substance. **Injection** can occur when high-pressure fluid penetrates skin. A small injection wound can conceal serious injury: report the event and obtain emergency medical care promptly under the emergency procedure. Never use a hand to search for a pressurized leak.

Hazards also act through noise, vibration, heat, cold and physical contact. Noise exposure needs assessment and controls; hearing protection is one part of the required response. Vibration can affect the hands and arms or the whole body, depending on the equipment and exposure. Heat risk depends on workload, weather, clothing and other conditions; a plan may include controls, acclimatization, suitable breaks, hydration and emergency response. Cold and wet conditions can affect health, dexterity and traction. A calendar month alone does not establish a safe exposure.

Biological hazards may include bacteria, other microorganisms or viruses encountered in the actual work environment. Contaminated water, waste or sharps require the relevant assessment and procedure. Workers should report an unexpected condition and keep others from exposure while the responsible person arranges a response. Do not handle an unknown sharp or contaminated material as an improvised classroom demonstration.

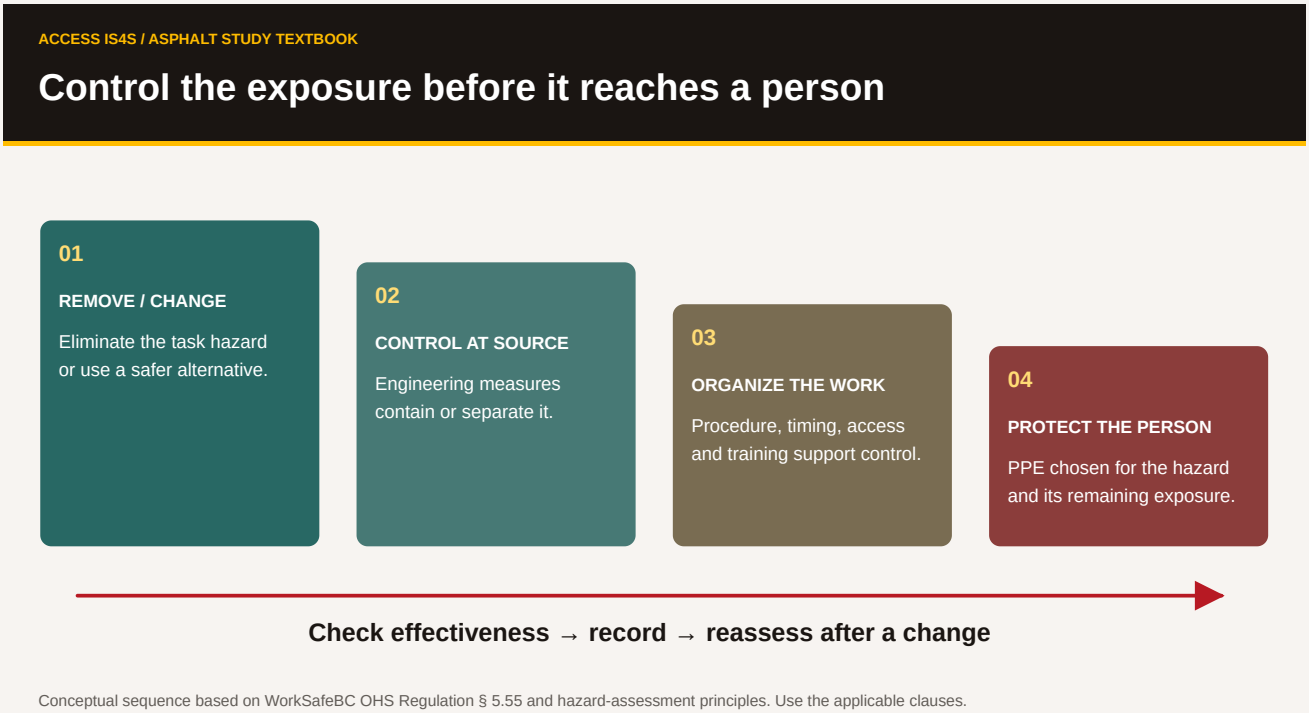
Musculoskeletal injury (MSI) involves muscles, tendons, joints or related tissues. Force, repetition, posture, duration and work organization can contribute. A shovel's load, a rake's reach and a wheelbarrow route affect the demand on a person. A change in tool, load handling, layout or work organization can reduce that demand. An awkward movement may cause a strain or sprain; a crush, impact or fall can cause trauma. Reporting early discomfort or fatigue helps the crew address a developing problem.

Put the control before the exposure

WorkSafeBC **5.55** requires hazardous-substance exposure to be eliminated or controlled below harmful levels and applicable exposure limits using the specified control approaches. A substitute must have known hazards and reduce the risk. PPE as the primary control is permitted only under the conditions in the clause; it is not an automatic first choice. Effective controls often work together.

FIGURE T21

Control the exposure before it reaches a person



Note. ACCESS IS4S. (2026). Control the exposure before it reaches a person [Original instructional SVG]. Based on WorkSafeBC. (2026 retrieved consolidation). Occupational Health and Safety Regulation, § 5.55. Conceptual teaching sequence; use the complete applicable clauses.

Worked example. A fictional crew proposes cutting near an occupied entrance. The observation “dust reaches the doorway” identifies a possible exposure route and affected people. It does not measure the substance or concentration. The next step is to prevent the uncontrolled exposure and have the responsible people assess the material, cutting method, controls and access arrangement. A useful record

identifies the task and location, what was observed, who may be exposed, the interim response, the person responsible and what must be verified before work resumes. Simply handing out unspecified masks leaves critical questions unanswered.

TB7-03 · Build a hazard-and-control record

Use four fictional task cards: (a) combustion equipment proposed beside an enclosed work space; (b) a worker eating with contaminated gloves; (c) repetitive shovelling along an obstructed route; (d) hot weather with a new worker and a changed workload. For each, identify the exposure mechanism, affected people, missing assessment information, a control question and the person responsible for resolving it. State what would trigger reassessment. **Skills:** Reading, Writing, Problem Solving.

TB7-04 · Reassess when the conditions change

Your group has prepared the cutting-task record from the worked example. The facility now needs its entrance for a scheduled delivery, and the wind direction has changed. Revise the record rather than carrying forward its previous conclusion. Create a visual task brief that another group can read in one minute; test whether they can identify the unresolved decisions. Each learner keeps one contribution and one revision. **Skills:** Adaptability, Creativity and Innovation, Collaboration, Communication.

7.3 · Orientation connects policy to a real workplace

A **company safety policy** states commitments and responsibilities. A **procedure** explains how a particular task is controlled. A **manual** may contain equipment or program instructions. **Orientation** connects a person to the workplace: who supervises them, what they are assigned, what hazards apply, what procedures they need and how to get help. Reading a general policy cannot replace a demonstration of the actual task.

WorkSafeBC **3.22–3.25** addresses young and new workers, including workplace-specific orientation and training before work begins, additional training when needed and records. Topics include rights and responsibilities, supervisor contact, hazards, safe work procedures, emergency procedures, first aid and instruction/demonstration of the work. A person experienced elsewhere can still be new to this workplace. Asking for more instruction when a task is unfamiliar is part of responsible work.

Fictional orientation card O-01. Learner: Rowan. Assigned task: document and tool identification under instructor supervision. Supervisor role: classroom instructor. Location: Cedar Learning Access practice drawing. Actual machinery operation: not assigned. First-aid contact, emergency location description and assembly point: to be confirmed for the real training venue. Permitted tools: inspected teaching samples selected by the instructor. Open items: venue emergency arrangements and demonstration boundaries. The card is incomplete until the real venue information is confirmed.

TB7-05 · Close the orientation gaps

Identify what O-01 does establish and what it leaves unresolved. Draft five precise orientation questions, including who supervises, which task is authorized, how help is reached, how a hazard is reported and how additional instruction is obtained. Rehearse a respectful stop-and-clarify conversation. Record the

questions and who must answer them; do not fill venue blanks with invented information. **Skills:** Reading, Writing, Communication.

7.4 · WHMIS and first aid are working systems

WHMIS connects hazard classification, labels, safety data sheets (SDSs), education and workplace training. A **supplier label** identifies a hazardous product and communicates prescribed hazard information. A **workplace label** is required in specified workplace circumstances and has its own information requirements. An **SDS** gives structured information about the product, hazards, handling, exposure controls, first aid and other matters. The label, SDS and actual product identifier must agree.

A pictogram is a starting point for understanding a hazard. It does not give a complete response procedure, exposure limit, glove selection or first-aid instruction. Some classified hazards do not require a pictogram. The environmental symbol may appear on an SDS or label but is not mandatory under WHMIS. Product classification is not determined by choosing whichever picture looks most alarming.

WHMIS SYMBOL STUDY

Read the pictogram with the product information



Flame
Flammability and related hazards



Flame over circle
Oxidizing hazard



Exploding bomb
Explosive / reactive hazards



Gas cylinder
Gases under pressure



Corrosion
Corrosive effects



Skull and crossbones
Severe acute toxicity



Health hazard
Serious health effects



Exclamation mark
Harmful / irritant effects



Biohazardous infectious materials
Infectious biological hazard



Environment
Aquatic hazard; not mandatory under WHMIS

Note. Official pictograms from Canadian Centre for Occupational Health and Safety. (2026). *WHMIS—Pictograms* and official symbol kit. Artwork reproduced unchanged. Short meanings are teaching summaries; classification and conditions require the actual product label/SDS. The environmental pictogram is not mandatory under WHMIS. Complete reference and provenance.

SDS section	What to locate in this chapter
1 · Identification	Product identifier, recommended use and supplier/emergency contact information
2 · Hazard identification	Classification, label elements and other hazards
4 · First-aid measures	Product-specific first-aid information and need for medical attention
6 · Accidental release measures	Spill precautions and response information
7 · Handling and storage	Safe handling, storage and incompatibility information

SDS section	What to locate in this chapter
8 · Exposure controls / personal protection	Applicable control and protection information
10 · Stability and reactivity	Conditions and incompatible materials to avoid

Fictional records exercise W-01. A container is marked “Training Cleaner A.” The available sheet is titled “Training Cleaner B,” revision 2. A colleague says the two are probably similar. No chemical composition, classification or use procedure is supplied in this exercise. The mismatch is the problem to solve; these invented names are not usable products or SDSs.

TB7-06 · Resolve a label-and-SDS mismatch

Explain why W-01 is insufficient for use. Identify the exact information needed and the action to prevent use while the responsible person resolves the mismatch. Use a device to open the local CCOHS label and SDS references from the WHMIS library, find the product-identifier requirement, and save/reopen a short record of what you found. Choose three pictograms from the figure, name their associated hazard meaning and explain why a picture alone cannot select a control. **Skills:** Reading, Digital, Writing, Problem Solving.

First aid depends on the workplace's assessed needs. WorkSafeBC **3.16** requires the employer's written assessment and appropriate equipment, supplies, facilities, attendants and services; related provisions address procedures and communication. A general training certificate does not establish that a particular site has the required coverage. The designated attendant needs the appropriate valid certification for the role, and workers need to know how to summon assistance. The written assessment is reviewed as required and when relevant circumstances change.

TB7-07 · Make the emergency message useful

For the real training venue, identify the confirmed first-aid contact, way to summon help, location description and person responsible for meeting responders. If any item is unknown, record who will confirm it before the activity. In a **spoken simulation only**, practise giving the location, nature of the incident, number of people involved and known immediate hazards. Do not place a real emergency call for practice. Explain why this classroom task does not issue a first-aid qualification. **Skills:** Communication, Writing, Collaboration.

7.5 · Rigging starts with qualified people and known information

Rigging connects and controls a load for lifting or handling using suitable equipment and arrangements. A **sling** connects the load to the lifting system. A **shackle**, hook or other fitting has an identified function, compatibility and working load limit. **Working load limit (WLL)** describes a permitted working load under stated conditions; it is not a universal number detached from configuration, angle or component condition.

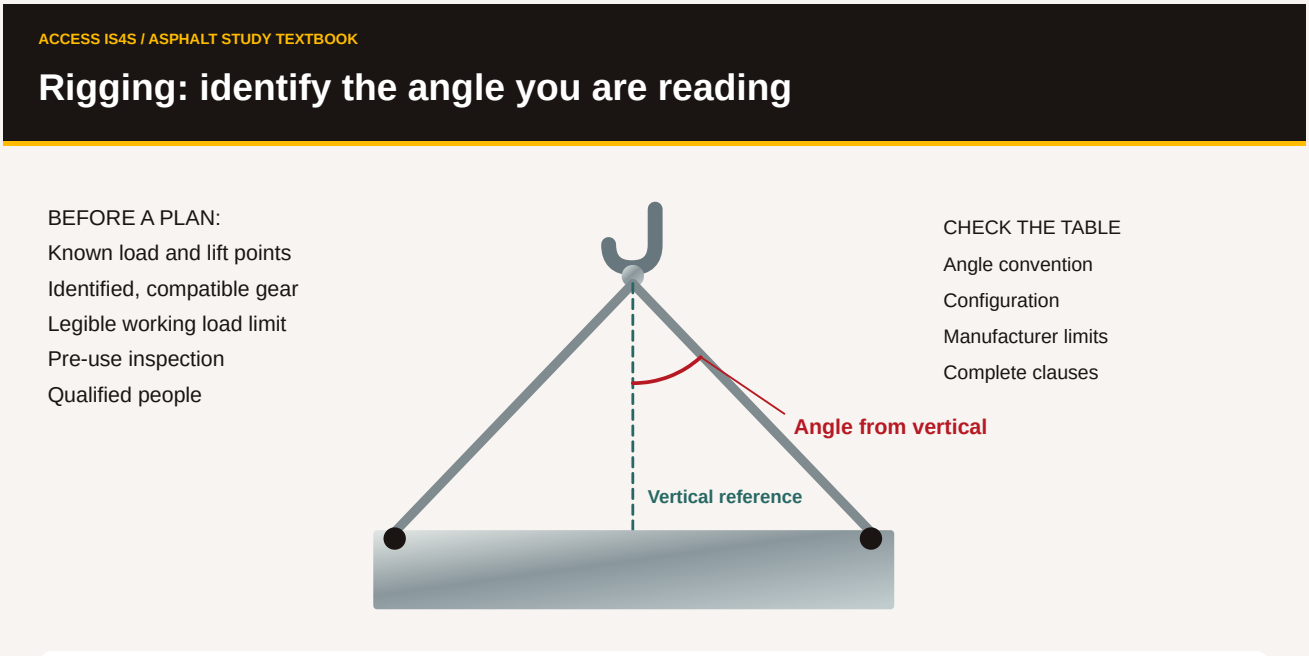
WorkSafeBC **Part 15** is the starting regulatory reading for this task. Under **15.2**, rigging and slinging must be done by or under the direct supervision of qualified workers familiar with the rigging and code of signals. The complete provisions address identification, design factors, WLL, operation, slings, components and

inspection. Manufacturer instructions and the actual lift planning are also needed. A learner must be able to recognize when the information or qualification is missing.

Reading checkpoint	Requirement or question to establish
Qualified person · 15.2	Who is qualified to rig and supervise the proposed work?
WLL and identification · 15.4–15.5	Is the equipment identified and its applicable WLL established?
Signals · 15.20	Which standard signals and communication arrangements apply?
Inspection · 15.31	Slings and attachments must be visually inspected before use on each shift; defective equipment is immediately removed from service.
Sling assembly · 15.33	The WLL of individual components and legs must not be exceeded under the specified rules.
Sling angle · 15.34	Angle is from vertical; required reduction is determined by a qualified person/manufacturer or the table as permitted.

FIGURE T22

Rigging: identify the angle you are reading



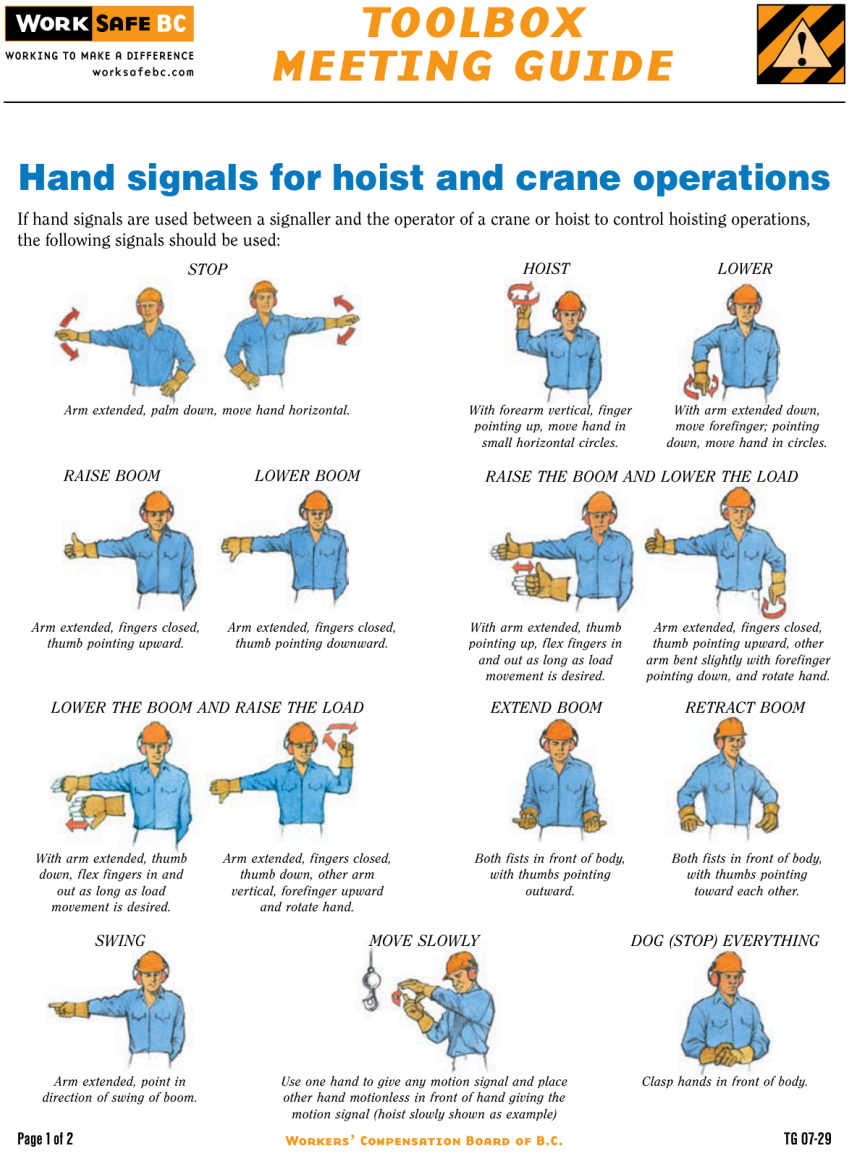
Note. ACCESS IS4S. (2026). Rigging: identify the angle you are reading [Original instructional SVG]. Based on WorkSafeBC. (2026 retrieved consolidation). Occupational Health and Safety Regulation, Part 15, especially §§ 15.2, 15.31 and 15.34. Identifies the angle from vertical; no working load limit or lift arrangement is prescribed.

An angle measured **from vertical** is different from an angle measured from horizontal. A reader must identify the convention before reading a table. The diagram shows the geometry, but supplies no load value, component identification or planned configuration. It cannot be used to select gear. Likewise, a label

on one leg does not establish the allowable load of an entire assembly.

FIGURE T23

Read the established crane signals



WORK SAFE BC
WORKING TO MAKE A DIFFERENCE
worksafebc.com

TOOLBOX MEETING GUIDE

Hand signals for hoist and crane operations

If hand signals are used between a signaller and the operator of a crane or hoist to control hoisting operations, the following signals should be used:

STOP
Arm extended, palm down, move hand horizontal.

HOIST
With forearm vertical, finger pointing up, move hand in small horizontal circles.

LOWER
With arm extended down, move forefinger; pointing down, move hand in circles.

RAISE BOOM
Arm extended, fingers closed, thumb pointing upward.

LOWER BOOM
Arm extended, fingers closed, thumb pointing downward.

RAISE THE BOOM AND LOWER THE LOAD
With arm extended, thumb pointing up, flex fingers in and out as long as load movement is desired.
Arm extended, fingers closed, thumb pointing upward, other arm bent slightly with forefinger pointing down, and rotate hand.

LOWER THE BOOM AND RAISE THE LOAD
With arm extended, thumb down, flex fingers in and out as long as load movement is desired.
Arm extended, fingers closed, thumb down, other arm vertical, forefinger upward and rotate hand.

EXTEND BOOM
Both fists in front of body, with thumbs pointing outward.

RETRACT BOOM
Both fists in front of body, with thumbs pointing toward each other.

SWING
Arm extended, point in direction of swing of boom.

MOVE SLOWLY
Use one hand to give any motion signal and place other hand motionless in front of hand giving the motion signal (hoist slowly shown as example)

DOG (STOP) EVERYTHING
Clasp hands in front of body.

Page 1 of 2
WORKERS' COMPENSATION BOARD OF B.C.
TG 07-29

Note. From Hand signals for hoist and crane operations (TG07-29, p. 1), by WorkSafeBC, n.d. Copyright WorkSafeBC. Complete first page reproduced. Compare the regulation's Figure 15-1 and the applicable lift plan. Source and context.

TB7-08 · Read a proposed lift for missing evidence

Fictional request: "Lift the crate with the slings beside it. It should be light enough. The tag on one sling cannot be read. We have not confirmed the load mass, lift points, configuration or qualified rigger." Write a response identifying at least five unresolved items. Explain why guessing the weight or using the clearest-looking sling is inadequate. Locate 15.2, 15.31 and 15.34 in the local regulation and record the requirements in your own words. No lift is performed. **Skills:** Reading, Writing, Problem Solving.

TB7-09 · Identify geometry and communicate clearly

Point to the vertical reference and sling leg in the diagram. Explain which angle Table 15-3 uses. Then, with the instructor, identify two illustrated crane signals on the official sheet and locate their names. Practise naming/recognizing the pictured signals in the classroom only; this exercise does not authorize signalling a crane. Explain how an unclear or conflicting signal would be handled through the actual safe-work procedure. **Skills:** Reading, Communication, Collaboration.

7.6 · Recognize the site and machine hazards

A paving machine has areas that move, heat, pivot or hold stored energy. A stopped engine does not prove all hazardous energy is controlled. Electrical energy, hydraulic pressure, elevated components, springs, gravity, hot surfaces and pressurized systems can remain hazardous. Isolation, lockout, blocking and verification require the applicable procedure and trained people. Do not reach into a feeder, auger or other moving mechanism to clear material.

FIGURE T24

Find the warning before the control



Illustration 5 g01409730

⚠ WARNING

Explosion Hazard! Improper jumper cable connections can cause an explosion resulting in serious injury or death. Batteries may be located in separate compartments. Refer to the Operation and Maintenance Manual for the correct jump starting procedure.

Hot Surface (4)

This safety message is located near the exhaust outlet.



Illustration 6 g03446436

⚠ WARNING

Hot engine components can cause injury from burns. Before performing maintenance on the engine, allow the engine and the components to cool.

Crush Hazard (5)

This safety message is on both sides of the machine near the articulation joint.

Note. Caterpillar. (2022). CB2.5 GC, CB2.7 GC, CC2.7 GC utility compactors: Operation and maintenance manual (M0130633-02, source PDF p. 9). Complete source-page render. Confirm the publication and serial-number applicability before operating. Source and context.

FIGURE T25

Articulation and the steering frame lock

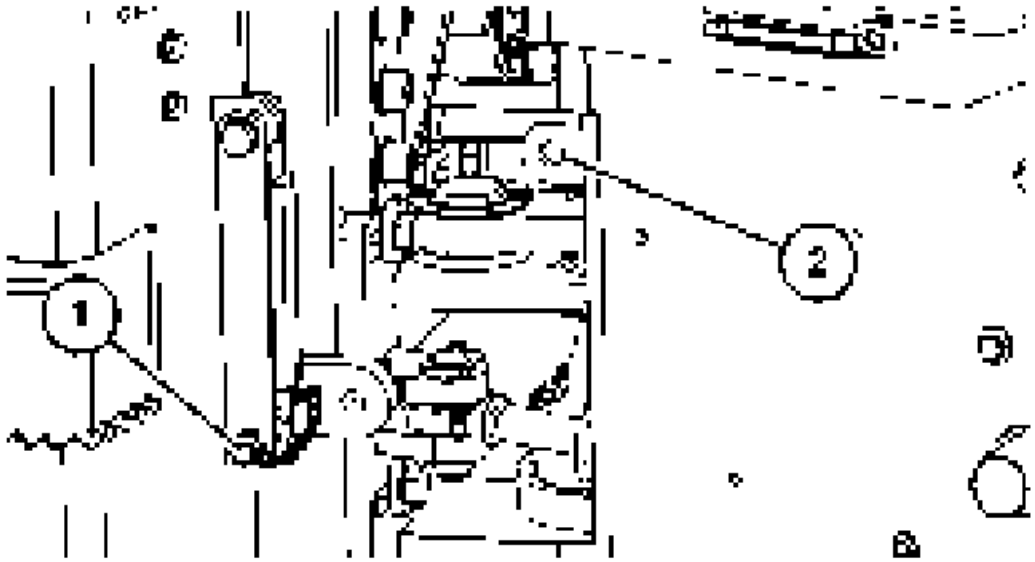


Illustration 34 g06183886

Unlocked position

Note. From CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022), source PDF p. 38), by Caterpillar, 2022. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Read the complete source page for locking/unlocking instructions; this task is identification only. Source and context.

The **articulation joint** allows sections of some machines to pivot. A **steering frame lock** is a model-specific protective device for the circumstances described in its manual. Its existence in a picture does not show that it has been installed or that all energy has been isolated. Batteries, tyres, hydraulic lines and fuel systems require their own procedures. Opening a hot or pressurized system, standing near a moving joint or relying on a raised component without the specified support can cause serious harm.

FIGURE T26

Locate all 19 original roller-control callouts

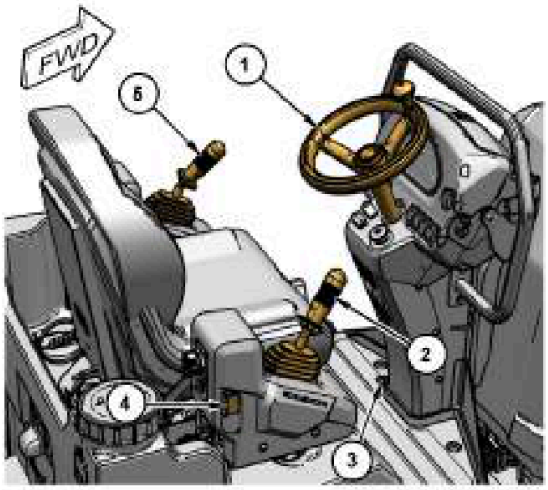


Illustration 39 g06673749

(1) Steering wheel
(2) Right propel control
(3) Traction Control Switch (if equipped)
(4) 12V power receptacle
(5) Left propel control (if equipped)

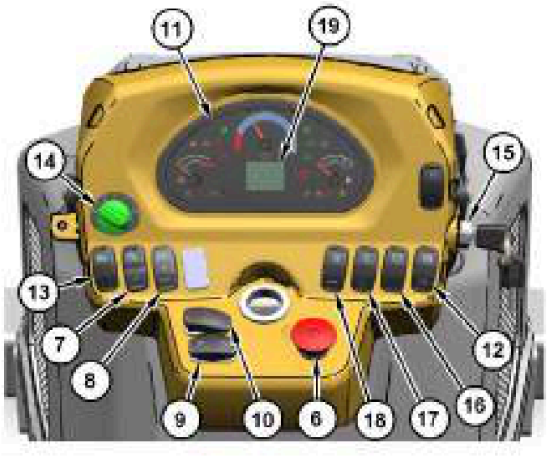


Illustration 40 g06610378

(6) Emergency stop knob
(7) Work lights
(8) Front/Position lights
(9) Turn signal
(10) Hazard lights
(11) Gauges and indicators
(12) Parking brake
(13) Horn
(14) Water spray system
(15) Engine start switch
(16) Vibratory mode / Emulsion switch
(17) Engine speed
(18) Beacon
(19) Digital display

Steering Wheel (1)

Steering wheel (1) controls the directional steering of the machine. The machine will turn in the same direction as steering wheel (1).

Note. From CB2.5 GC, CB2.7 GC, CC2.7 GC Utility Compactors — Operation and Maintenance Manual (M0130633-02 (August 2022), source PDF p. 41), by Caterpillar, 2022. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: CB2.5 GC: XGC 1-UP, 640 1-UP; CB2.7 GC: BT7 1-UP, 641 1-UP; CC2.7 GC: 2C7 1-UP, 642 1-UP. Illustration 39 has 1–5; Illustration 40 has 6–19. Optional items retain "if equipped." Source and context.

Look and name. A control panel illustrates why training must match the model. Labels, interlocks and functions vary. Identify the words that you can read in this manual figure, then identify which publication and machine applicability you would verify before touching a real control. The picture is not a starting or shutdown sequence.

Site risk	What must be established before the work
Slips, trips and falls	Access surface, housekeeping, lighting, edges and the applicable fall-protection arrangement

Site risk	What must be established before the work
Moving vehicles, reversing and crush points	Traffic arrangement, visibility, pedestrian separation and agreed communication
Hot mixture, surfaces and burns	Exposure points, separation, suitable protection and emergency response
Overhead utilities	Actual line/voltage information, required limits of approach and the applicable authority's procedure; no single clearance fits every case
Underground utilities	BC 1 Call request and responses, affected owners, verified location and safe excavation procedure; a web-map line is insufficient
Noise and vibration	Assessed exposure, equipment/work controls and required protection/program
Fatigue, illness, drugs or alcohol impairment	Fitness for the task, reporting/support arrangements and reassignment where needed; pressure to continue does not remove the hazard
Inexperience, inattention or complacency	Task-specific instruction, supervision, clear limits and renewed checks after a change

A public utility map can help ask a question. It does not establish a buried asset's exact position, depth, status or clearance for excavation. A record's owner, maintenance responsibility and operational responsibility may be different fields. A record marked proposed or abandoned must not be silently treated as a verified in-service line. Chapter 5's map desk teaches these distinctions with fictional data.

TB7-10 · Use the manual to ask the right question

In the roller safety page, find one warning and identify the hazard and stated response. In the steering-frame illustration, identify the lock and the joint it relates to. State the publication/model context, what the image establishes and what must be verified on the actual machine. Explain why a key switched off or a visible lock is insufficient evidence that every energy source is controlled. **Skills:** Reading, Problem Solving, Writing.

TB7-11 · Stop an unsupported site assumption

Fictional messages: “The utility line on the map is exact”; “The overhead wire looks high enough”; “The engine is off, so the raised part is safe”; “We've done this for years, so no new check is needed.” Rewrite each as a precise verification question and an appropriate interim response that avoids exposure. For one case, use the plan desk and save a note identifying the source status and the missing information. **Skills:** Adaptability, Digital, Writing, Problem Solving.

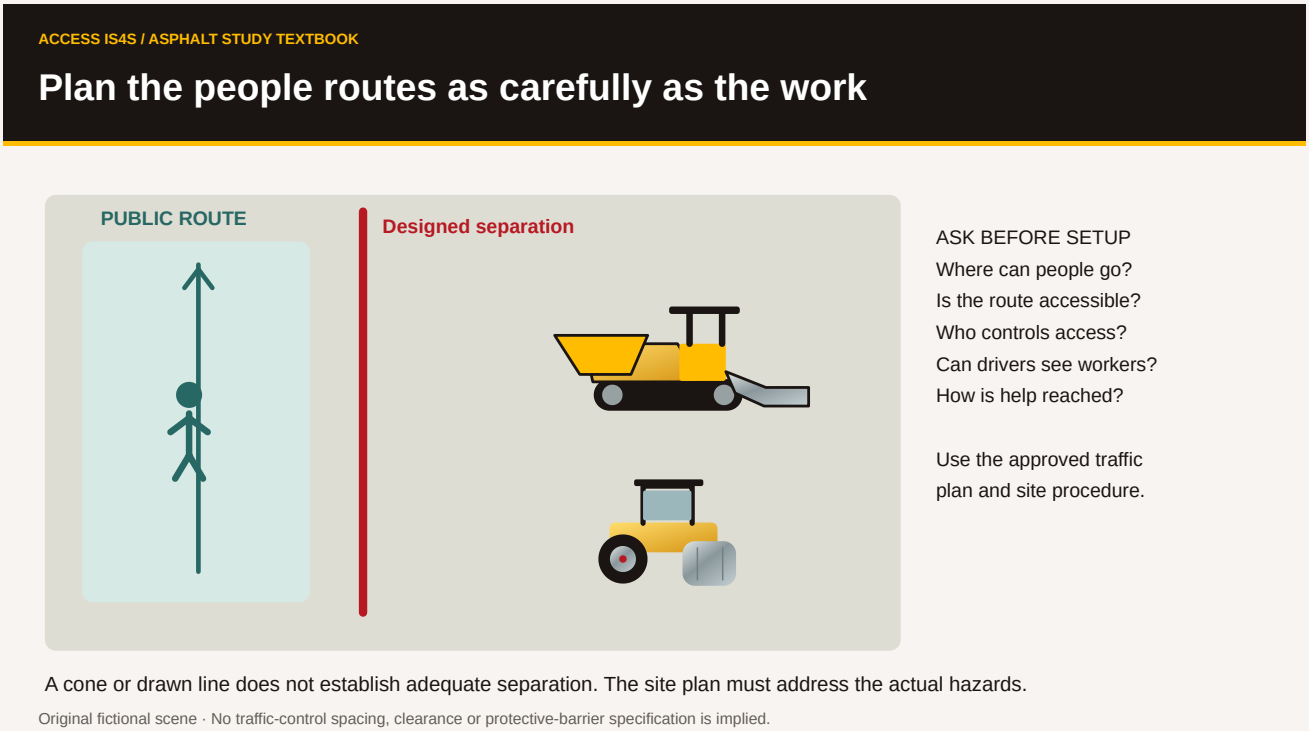
7.7 · Protect the public and secure the equipment

A site can affect people who never joined the crew briefing: visitors, delivery drivers, people using mobility aids, nearby workers and people passing the entrance. The work plan must address how they move, how they are kept from hazards and how information reaches them. Traffic control follows the applicable road

authority's plan and WorkSafeBC Part 18 requirements. A hand-drawn arrow or cone in a teaching picture does not establish a compliant layout.

FIGURE T27

Plan the people routes as carefully as the work



Note. ACCESS IS4S. (2026). Plan the people routes as carefully as the work [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

At the end of a work period, the crew still has responsibilities. Equipment must be parked and secured under the actual manufacturer and site procedures. Brakes, wheel chocks where required, isolation and the treatment of raised components depend on the equipment and conditions. Remove access to unauthorized operation and verify the planned public/site boundaries. A generic instruction to “turn it off” leaves stored energy, movement and access questions unresolved.

Worked example. A crew has secured a machine under its procedure, but the public route now passes beside a hot component. Completing one equipment-security task has not resolved the public-exposure hazard. The responsible people must reassess the access arrangement, control the exposure and communicate the changed route. The handover records both the machine status and the remaining site condition.

TB7-12 · Review the public route

Use the conceptual diagram. Identify who may need access, how a pedestrian/vehicle conflict could arise and what the drawing cannot establish. Create five questions for the responsible site planner, including accessibility, visibility, control of entry, emergency access and changing conditions. Improve the clarity of the public-route notice without inventing a traffic-control layout or barrier specification. **Skills:** Communication, Creativity and Innovation, Problem Solving.

TB7-13 · Integrated classroom safety brief

Prepare a two-page brief for the fictional Cedar Learning Access site. Include: task and limits; PPE decision record; four hazard/control questions; orientation gaps; label/SDS match check; first-aid contact verification; rigging prerequisites; utility information limits; and public/equipment-security handover. Reuse your earlier work but resolve contradictions. List the responsible role and needed evidence beside each open item. Attach your checked Chapter 5 quantity/revision sheet and the angle-convention diagram from TB7-09. Add a visual that helps a classmate find a critical question quickly.

Exchange briefs. Your partner tests whether they can find the unresolved issues without your explanation. Revise, save and reopen your final brief. Each learner presents one decision and one change they made after feedback. **Skills:** all nine Skills for Success. The evidence is your accurate reasoning and clear record, including the points that still require qualified or site-specific confirmation.

Working reference and evidence record

Words used in this book

Term	Meaning in these lessons
Aggregate	Mineral material such as sand, gravel or crushed stone
Asphalt binder / asphalt cement	Binding material in an asphalt mixture; distinct from Portland cement
Batch plant	Production arrangement that makes identifiable batches
Drum-mix plant	Production arrangement using a continuous material/mixing process; layouts vary
RAP	Reclaimed asphalt pavement material considered for an approved reuse pathway
Hot-in-place recycling	Specialist process treating existing asphalt pavement on the road
Screed	Paver component that contributes to forming the placed mat
Compaction	Densification process performed according to the applicable approved procedure
Milling	Removal or profiling of existing material using suitable equipment
Chainage	Position measured along a defined project line
Plan / profile / cross-section	Top view / elevation along a line / cut across a route
Crown	Raised transverse road form that can direct surface water toward sides
Superelevation	Cross-slope arrangement associated with a curve; design is project-specific
Runoff	Water moving from a surface
Specification	Documented requirements; applicability depends on its legal/contractual context
Special provision	Project/contract-specific provision that may modify or add requirements
Revision	Identified version of a document
Functional comparison	Comparison based on an equivalent required service, period and conditions
Life-cycle boundary	Stages included and excluded in an environmental assessment
Percentage point	Difference between two percentages expressed on the same basis
EPS in the 2006 specifications	End Product Specification, explicitly defined in clause 502.03.01; identify the edition and context before applying the term
EPS geofoam	Expanded-polystyrene material for an engineered geotechnical application; an abbreviation's meaning still depends on context

Read the word, find the evidence, explain the limit

To practise a term, write the trade word, point to its feature or source statement, and explain the function in one plain sentence. Then say what the picture or statement cannot establish. Keep technical words intact while making the explanation easier to follow. For example: “The **asphalt course** is the surface layer shown here; this picture does not specify its required thickness.”

Record evidence without overstating it

For each selected activity, record: activity ID; date; task and document version; what you did; support used; result; check; feedback; revision; and next practice. Mark the observed skill. A group product needs an individual contribution record. An accurate calculation needs units and assumptions. A digital claim needs actual tool use. A future application is a plan until it has been observed.

Activity	Evidence kept	Support and feedback	Next practice
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Sources and continuing questions

The research register provides public titles, issuers, dates, URLs, retained short excerpts and retrieval hashes. It distinguishes dated factual evidence from invented classroom inputs and records questions that need current local or technical confirmation. Use the register to check a claim, then keep the question and the evidence together. A link or a confident statement alone is not the complete answer.

Supplied technical document

British Columbia Ministry of Transportation, Construction and Maintenance Branch. *2006 Design Build Standard Specifications for Highway Construction*, adopted 20 February 2006. Section 502.03.01, PDF 297 / printed 1 of 22, defines End Product Specification (EPS). The supplied historical document is distinct from the current public specifications page below; its presence does not establish current contract applicability.

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The curated evidence register gives the teaching use and limits; exact selected excerpts and hashes support checking. The road-industry table extract preserves the nine source rows.

Trade-outline and figure references (APA 7)

SkilledTradesBC. (2016). *Asphalt paving/laydown technician program outline* (August 2016). Program outline, Line A, printed pp. 20–25 / PDF pp. 22–27.

ACCESS IS4S. (2026a). *Follow the material, step by step* [SVG instructional illustration]. Local illustration.

ACCESS IS4S. (2026b). *See how the layers work together* [SVG instructional illustration]. Local illustration.

Figures LA-1 and LA-2 are original ACCESS IS4S teaching illustrations. Figure LA-3 retains the manufacturer's component callouts and is accompanied by a selected text legend. The trade terms are connected to the supplied outline; source-specific technical claims retain their evidence cards and reference entries.